Primes in Arithmetic Progressions

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Contents

Euler's Idea on Primes

Theorem (Euler).

The series $\sum_q \frac{1}{q}$ diverges, where q ranges over the prime numbers.

- Infinitude of Primes
- Density of Primes
- Motivation is important!

Euler's Idea on Primes

Proof. (Sketch)

The Euler product formula: for $s > 1$,

$$\zeta(s) = \sum_n \frac{1}{n^s} = \prod_q \left(1 + \frac{1}{q^s} + \frac{1}{q^{2s}} + \cdots\right) = \prod_q \left(1 - \frac{1}{q^s}\right)^{-1}.$$

Take logarithms of both sides:

$$\log \sum_n \frac{1}{n^s} = -\sum_q \log\left(1 - \frac{1}{q^s}\right) = \sum_q \frac{1}{q^s} + O(1).$$

As $s \to 1^+$, the left-hand side tends to $+\infty$, so

$$\sum_q \frac{1}{q^s} \to \infty.$$

Arithmetic Progressions

Definition (Arithmetic Progression).

An *arithmetic progression* is a finite or infinite list of numbers of the form

$$a,\ a+q,\ a+2q,\ \ldots$$

where a and q are fixed integers. The number q is called the *common difference*.

Main Question

Are there infinitely many primes in **arithmetic progressions**?

- Birth of **Analytic Number Theory (1837)**.

Dirichlet's Dream

Fix integers m and k with $\gcd(m, k) = 1$, and let $s > 1$ be a real number. We wish to show

$$\sum_{q \equiv m \bmod k} \frac{1}{q^s}$$

diverges.
Write this as

$$\sum_{q} \frac{1_{m,k}(q)}{q^s}$$

where

$$1_{m,k}(q) = \begin{cases} 1 & \text{if } q \equiv m \bmod k, \\ 0 & \text{otherwise,} \end{cases}$$

and try to repeat Euler's argument.

Dirichlet's Theorem on Primes in Arithmetic Progressions

Theorem (Dirichlet).

If $q > 0$ and $(a, q) = 1$ there are infinitely many primes in the arithmetic progression $a + nq, \; n = 0, \; 1, \; 2, \; \ldots$

Let us formalize Dirichlet's idea via **Dirichlet characters** and the celebrated **Dirichlet L-function** $L(s, \chi)$.

Dirichlet Character (I)

Definition (Dirichlet Character).

Let $k \geq 1$ be an integer. A **Dirichlet character modulo** k is an arithmetic function

$$\chi : \mathbb{Z} \to \mathbb{C}$$

satisfying the properties:

- Periodicity modulo k:

$$\chi(n + k) = \chi(n) \quad \text{for all} \quad n \in \mathbb{Z}.$$

- Completely multiplicative:

$$\chi(mn) = \chi(m)\chi(n) \quad \text{for all} \quad m, n \in \mathbb{Z}.$$

- Vanishing on non-units:

$$\chi(n) = 0 \iff \gcd(n, k) > 1.$$

Remark.

Equivalently, a Dirichlet character modulo k is a group homomorphism

$$\chi : (\mathbb{Z}/k\mathbb{Z})^{\times} \to \mathbb{C}^{\times},$$

extended to all integers by setting $\chi(n) = 0$ whenever $\gcd(n, k) > 1$.

Definition (Principal Character).

A Dirichlet character χ is called **principal** and denoted by χ_1 if

$$\chi_1(n) = \begin{cases} 1 & \text{if } \gcd(n, k) = 1, \\ 0 & \text{if } \gcd(n, k) > 1. \end{cases}$$

Dirichlet Character (III)

Proposition.

We have $\varphi(k)$ many Dirichlet characters modulo k, where φ denotes Euler's phi function.

The set of characters on $(\mathbb{Z}/k\mathbb{Z})^\times$ forms a group $\widehat{(\mathbb{Z}/k\mathbb{Z})}^\times$ with pointwise multiplication, identity χ_1, and inverse elements $\overline{\chi}$. In fact, $|(\mathbb{Z}/k\mathbb{Z})^\times| = |\widehat{(\mathbb{Z}/k\mathbb{Z})}^\times| = \varphi(k)$.

Theorem (Orthogonality of Dirichlet Characters).

Let $\chi_1, \ldots, \chi_{\varphi(k)}$ denote the $\varphi(k)$ Dirichlet characters modulo k. Let m and n be two integers with $\gcd(n, k) = 1$. Then we have

$$\sum_{r=1}^{\varphi(k)} \chi_r(m)\,\overline{\chi_r(n)} = \begin{cases} \varphi(k) & \text{if } m \equiv n \pmod{k}, \\ 0 & \text{if } m \not\equiv n \pmod{k}. \end{cases}$$

Dirichlet L-Function

Definition (Dirichlet L-Function).

Let χ be a Dirichlet character and $s = \sigma + it$ a complex variable with $\sigma > 1$. The **Dirichlet L-series** is defined as:

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

With analytic continuation, it can be extended to an analytic function on $\mathbb{C}$; it is then called a **Dirichlet L-function**.

Dirichlet's Theorem on Primes in APs: Proof Idea (I)

Euler Product Formula for $L(s, \chi)$

Similar to $\zeta(s)$, the L-function has an Euler product:

$$L(s, \chi) = \prod_{p} \frac{1}{1 - \chi(p)p^{-s}} \quad \text{for} \quad \sigma > 1,$$

where p is prime. For the principal character χ_1, $L(s, \chi_1)$ reduces to $\zeta(s)$, which has a simple pole at $s = 1$.

We can write the Euler product expansion as:

$$L(s, \chi) = \prod_{q} \left(1 - \frac{\chi(q)}{q^s}\right)^{-1} = \prod_{q \nmid k} \left(1 - \frac{\chi(q)}{q^s}\right)^{-1}.$$

Dirichlet's Theorem on Primes in APs: Proof Idea (II)

Taking logarithms of both sides yields

$$\log L(s,\chi) = \sum_{q \nmid k} \frac{\chi(q)}{q^s} + O(1).$$

Multiply both sides by $\overline{\chi(m)}$ and sum over χ:

$$\sum_{\chi} \overline{\chi(m)} \log L(s,\chi) = \sum_{\chi} \sum_{q \nmid k} \overline{\chi(m)} \frac{\chi(q)}{q^s} + O(1).$$

Using the orthogonality relations,

$$\sum_{\chi} \overline{\chi(m)} \log L(s,\chi) = \varphi(k) \sum_{q \equiv m \pmod{k}} \frac{1}{q^s} + O(1).$$

Dirichlet's Theorem on Primes in APs: Proof Idea (III)

$$\frac{1}{\varphi(k)} \sum_{\chi} \overline{\chi(m)} \log L(s,\chi) = \sum_{q \equiv m \pmod{k}} \frac{1}{q^s} + O(1).$$

Let $s \to 1^+$.

We divide the characters into three types:

1. The principal character χ_1.
2. The non-principal real-valued characters.
3. The complex characters.

Show:

- $L(s,\chi_1)$ has a simple pole at $s = 1$.
- Non-vanishing of $L(1,\chi)$ for $\chi \neq \chi_1$.

Non-Vanishing of $L(1, \chi)$

Theorem (Non-vanishing at $s = 1$).

Let χ be a Dirichlet character modulo k.

- If $\chi \neq \chi_1$ is non-principal, then $L(s, \chi)$ is holomorphic at $s = 1$ and
$$L(1, \chi) \neq 0.$$

- For complex characters, one proves that certain positive expressions built from $|L(1, \chi)|^2$ stay bounded away from 0, forcing $L(1, \chi) \neq 0$.
- For real non-principal characters, one constructs a hyperbolic sum whose asymptotics imply $L(1, \chi) > 0$, ruling out $L(1, \chi) = 0$.
- This non-vanishing is the key input that forces the sum
$$\sum_{q \equiv m \pmod{k}} \frac{1}{q^s}$$
to diverge as $s \to 1^+$, completing the proof of Dirichlet's theorem.

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